

WEEK 4 PROJECT SUMMARY

Problem Addressed

HealthConnect Clinic is seeking to reduce missed patient appointments and improve the patient support experience. The analysis focuses on understanding appointment attendance and identifying factors that may be associated with patient no-shows.

Resources Used

The analysis uses the provided `HealthConnect_Appointment_Data.csv` dataset, supported by the project brief and available HealthConnect project resources. Python and Jupyter Notebook are used for data inspection, validation, and initial analysis.

Key Observations

The dataset contains 5,000 appointment records across 18 variables and 1,696 unique patients. The initial assessment identified 2,423 No-Show appointments (48.46%), 2,314 Attended appointments (46.28%), and 263 Cancelled appointments (5.26%). No duplicate records were identified. Missing values were observed in `reminder_channel`, `distance_to_clinic_km`, and `waiting_time_minutes`.

Initial exploration indicates that previous no-show behaviour, reminder activity, appointment characteristics, booking lead time, distance to the clinic, and waiting time warrant further investigation as potential factors associated with appointment attendance.

Proposed Approach

The next phase will conduct deeper exploratory analysis across patient, appointment, booking, reminder, and accessibility-related variables. The analysis will develop and evaluate relevant KPIs, identify meaningful patterns in no-show behaviour, and translate the findings into practical business insights and recommendations.

Key Considerations

The analysis will distinguish between association and causation, particularly when assessing reminder effectiveness. Missing values and small subgroups will be handled carefully, while the original dataset will remain unchanged.

Week 5 Focus

Week 5 will focus on deeper data analysis and identifying the strongest patterns associated with missed appointments. The findings will provide the evidence base for developing actionable recommendations to support improved appointment attendance and patient service delivery.